CAS SciFinder

Task History

Initiating Search

November 6, 2025, 4:58 PM

Search:

CAS Lexicon Search:

Sodium triacetoxyborohydride - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (1,755)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,353) Filtered By: Yield: **90-100%, 80-89%, 70-79%, 50-69%** Document Type: **Journal** Language: **English**	Reactions	View Results

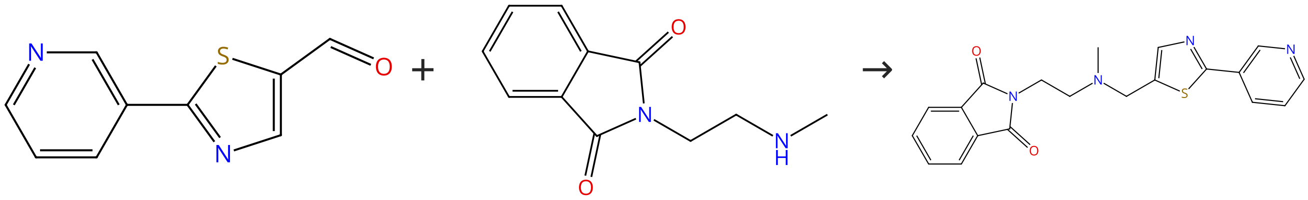
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (28)

Suppliers (22)

31-313-CAS-4645117	Steps: **1** Yield: **100%**
1.1	**Reagents:** Acetic acid, 5-(3-Pyridinyl)-2-thiophenecarboxaldehyde **Solvents:** Dichloromethane; 4 h, rt
1.2	**Reagents:** Sodium bicarbonate **Solvents:** Water; rt
Experimental Protocols	

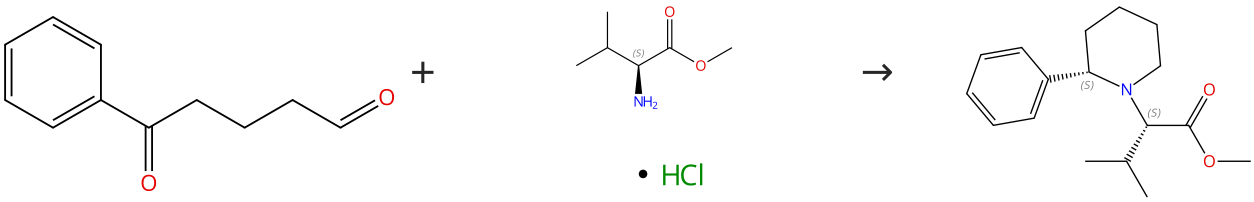
Synthesis and Antibacterial Activity of Novel C_{12} Vinyl Ketolides

By: Burger, Matthew T.; et al

Journal of Medicinal Chemistry (2006), 49(5), 1730-1743.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (9)

Absolute stereochemistry shown, Rotation (+)

Suppliers (92)

Absolute stereochemistry shown

Supplier (1)

31-313-CAS-10073954	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium triacetoxyborohydride **Solvents:** Dichloromethane

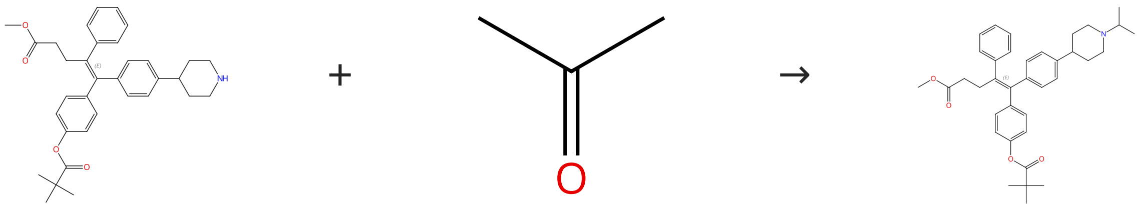
Reductive amination of 1,4- and 1,5-dicarbonyl compounds with (S)-valine methyl ester. Synthesis of (S)-2-phenylpyrrolidine and (S)-2-phenylpiperidine

By: Manescalchi, Francesco; et al

Tetrahedron Letters (1994), 35(17), 2775-8.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Double bond geometry shown

Suppliers (296)

Double bond geometry shown

31-313-CAS-23389345 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane; 0 °C; 12 h, 60 °C

1.2 **Reagents:** Water

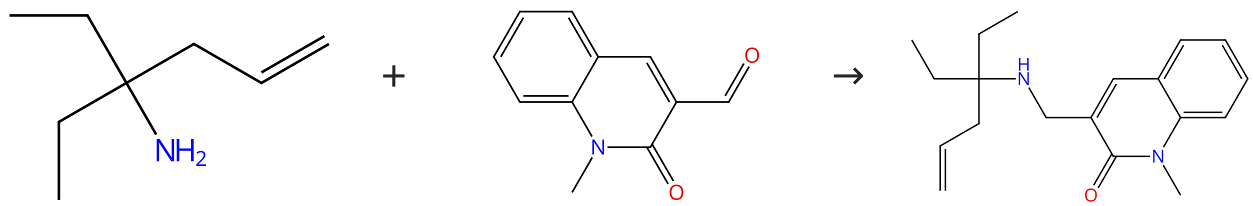
Experimental Protocols

An orally available inverse agonist of estrogen-related receptor gamma showed expanded efficacy for the radioi odine therapy of poorly differentiated thyroid cancer

By: Kim, Jina; et al

European Journal of Medicinal Chemistry (2020), 205, 112501.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (4) Suppliers (61)

31-313-CAS-6245727 Steps: **1** Yield: **100%**

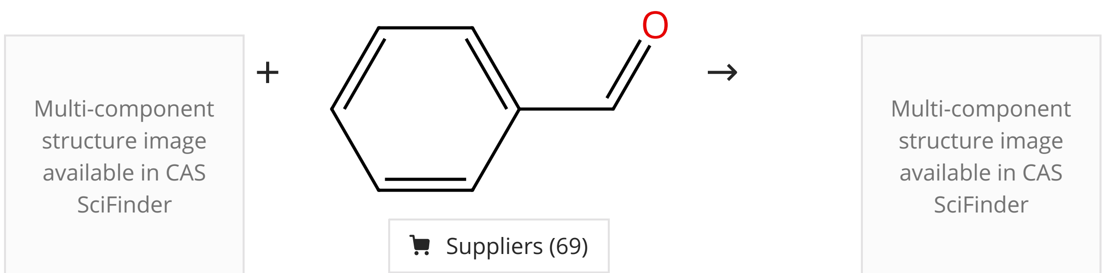
1.1 **Reagents:** Sodium triacetoxyborohydride; rt

Synthesis and evaluation of a new series of Neurop eptide S receptor antagonists

By: Melamed, Jeffrey Y.; et al

Bioorganic & Medicinal Chemistry Letters (2010), 20(15), 4700-4703.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (69)

31-313-CAS-3024062 Steps: **1** Yield: **100%**

1.1 **Reagents:** Borate(1-), tris(benzoato-κ*O*)hydro-, sodium (1:1), (*T*-4)-
Solvents: *N*-Methyl-2-pyrrolidone; 16 h, 70 °C

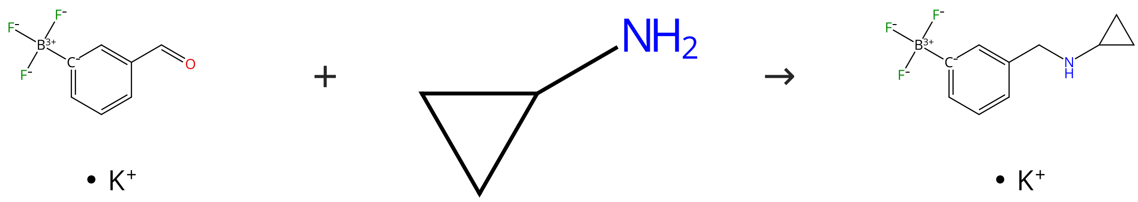
Experimental Protocols

Selective Transformation of a Crown Ether/sec-Ammonium Salt-Type Rotaxane to N-Alkylated Rotaxanes

By: Suzuki, Sakiko; et al

Organic Letters (2010), 12(4), 712-715.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (51) Suppliers (69) Suppliers (3)

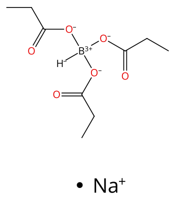
31-313-CAS-5956155	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium bifluoride **Solvents:** Methanol; 2 h, rt
1.2	**Reagents:** Pyridine-borane; 16 h, rt
Experimental Protocols	

Functionalization of Organotrifluoroborates: Reductive Amination

By: Molander, Gary A.; et al

Journal of Organic Chemistry (2008), 73(10), 3885-3891.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Supplier (1)

+

Multi-component structure image available in CAS SciFinder

→

Multi-component structure image available in CAS SciFinder

31-313-CAS-622445	Steps: **1** Yield: **100%**
1.1	**Solvents:** *N*-Methyl-2-pyrrolidone; 16 h, 70 °C
Experimental Protocols	

Selective Transformation of a Crown Ether/sec-Ammonium Salt-Type Rotaxane to N-Alkylated Rotaxanes

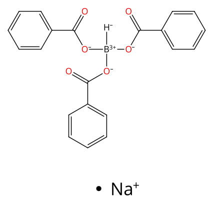
By: Suzuki, Sakiko; et al

Organic Letters (2010), 12(4), 712-715.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**

Multi-component structure image available in CAS SciFinder

+



→

Multi-component structure image available in CAS SciFinder

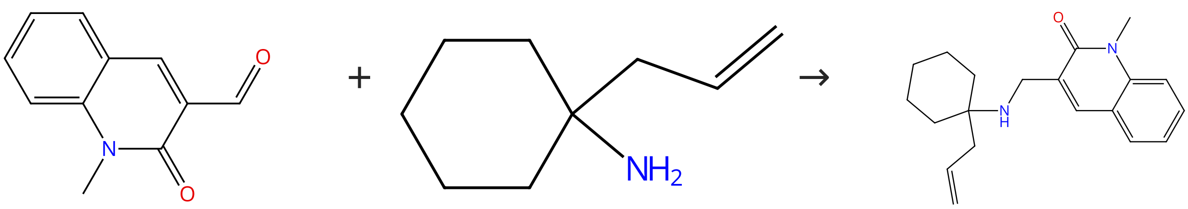
31-313-CAS-912064	Steps: **1** Yield: **100%**
1.1	**Solvents:** *N*-Methyl-2-pyrrolidone; 16 h, 70 °C
Experimental Protocols	

Selective Transformation of a Crown Ether/sec-Ammonium Salt-Type Rotaxane to N-Alkylated Rotaxanes

By: Suzuki, Sakiko; et al

Organic Letters (2010), 12(4), 712-715.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (61) Suppliers (25)

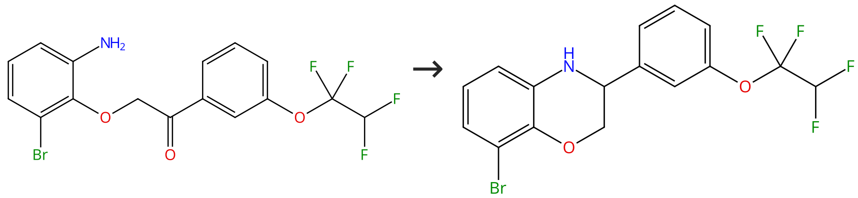
31-313-CAS-8384349	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium triacetoxyborohydride; rt

Synthesis and evaluation of a new series of Neurop eptide S receptor antagonists

By: Melamed, Jeffrey Y.; et al

Bioorganic & Medicinal Chemistry Letters (2010), 20(15), 4700-4703.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



31-313-CAS-2220817 Steps: **1** Yield: **100%**

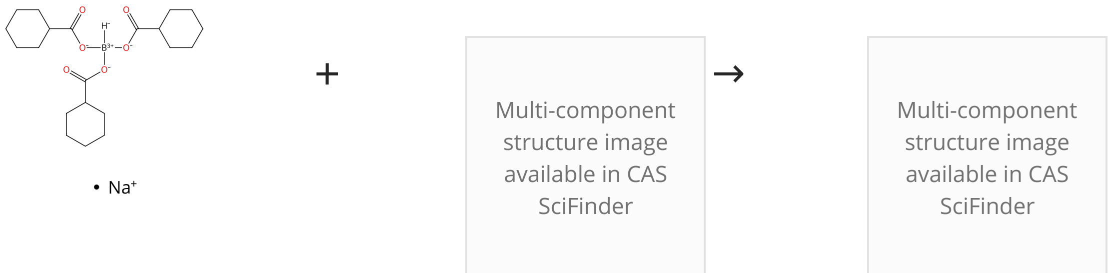
1.1 **Reagents:** Acetic acid, Sodium triacetoxyborohydride
Solvents: Dichloromethane; rt

Synthesis and discovery of 2,3-dihydro-3,8-diphenylbenzo[1,4]oxazines as a novel class of potent cholesteryl ester transfer protein inhibitors

By: Wang, Aihua; et al

Bioorganic & Medicinal Chemistry Letters (2010), 20(4), 1432-1435.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



31-313-CAS-12178398 Steps: **1** Yield: **100%**

1.1 **Solvents:** *N*-Methyl-2-pyrrolidone; 16 h, 70 °C

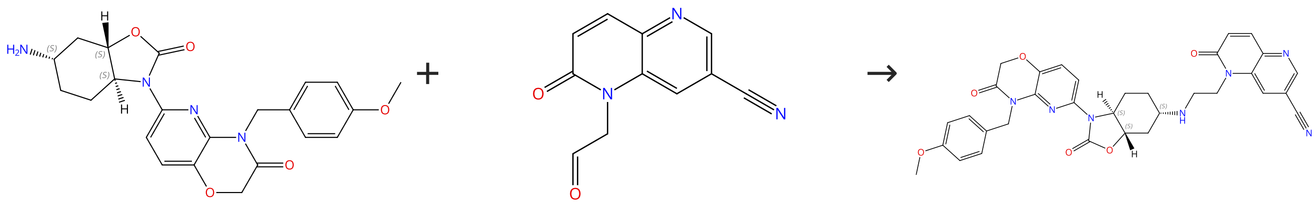
Experimental Protocols

Selective Transformation of a Crown Ether/sec-Ammonium Salt-Type Rotaxane to N-Alkylated Rotaxanes

By: Suzuki, Sakiko; et al

Organic Letters (2010), 12(4), 712-715.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Suppliers (2)

Absolute stereochemistry shown

31-614-CAS-34196351 Steps: **1** Yield: **100%**

1.1 **Solvents:** Dichloromethane; 2 h, rt

1.2 **Reagents:** Sodium triacetoxyborohydride; 17 h, rt

Experimental Protocols

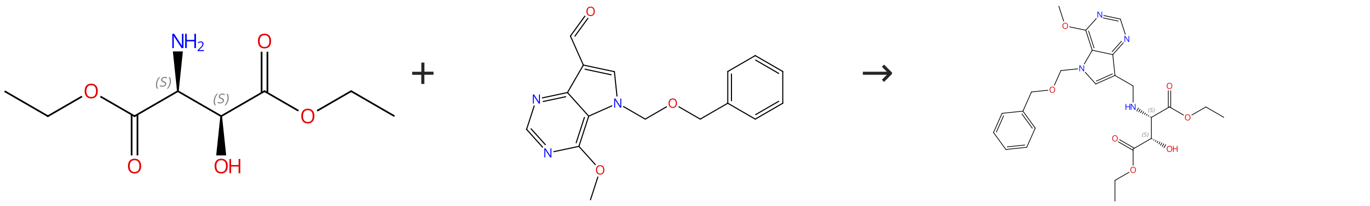
Discovery and structure-activity relationships of a novel oxazolidinone class of bacterial type II topoisomerase inhibitors

By: Lyons, Amanda; et al

Bioorganic & Medicinal Chemistry Letters (2022), 65, 128648.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (+)

Supplier (1)

Supplier (1)

Absolute stereochemistry shown

Supplier (1)

31-313-CAS-7267062	Steps: **1** Yield: **100%**
1.1	**Reagents:** Acetic acid, Sodium cyanoborohydride **Solvents:** Methanol; 18 h, rt
Experimental Protocols	

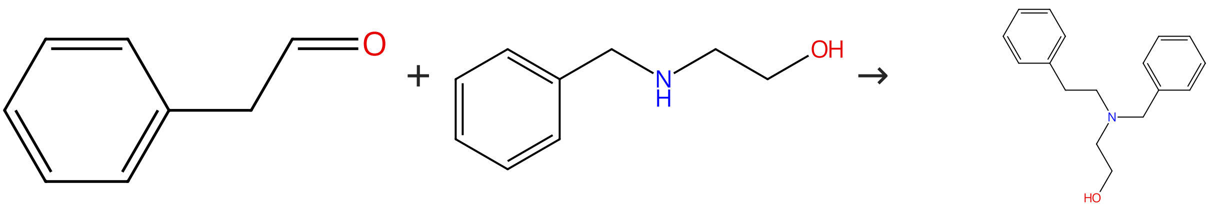
Third-Generation Immucillins: Syntheses and Bioactivities of Acyclic Immucillin Inhibitors of Human Purine Nucleoside Phosphorylase

By: Clinch, Keith; et al

Journal of Medicinal Chemistry (2009), 52(4), 1126-1143.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (69)

Suppliers (89)

Suppliers (9)

31-313-CAS-4398635	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium triacetoxyborohydride **Solvents:** 1,2-Dichloroethane; 14 h, rt
Experimental Protocols	

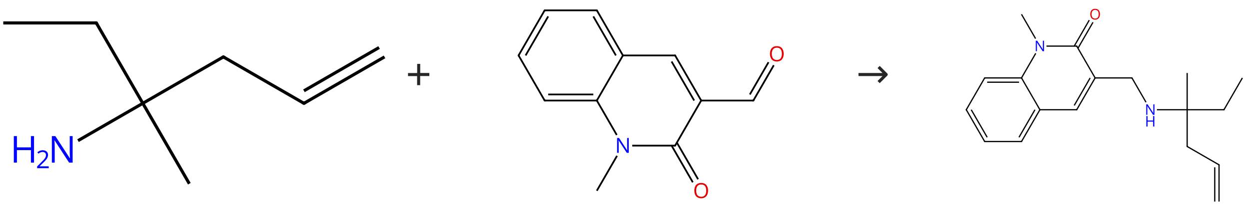
Discovery of Highly Potent and Selective Inhibitors of Neuronal Nitric Oxide Synthase by Fragment Hopping

By: Ji, Haitao; et al

Journal of Medicinal Chemistry (2009), 52(3), 779-797.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (9)

Suppliers (61)

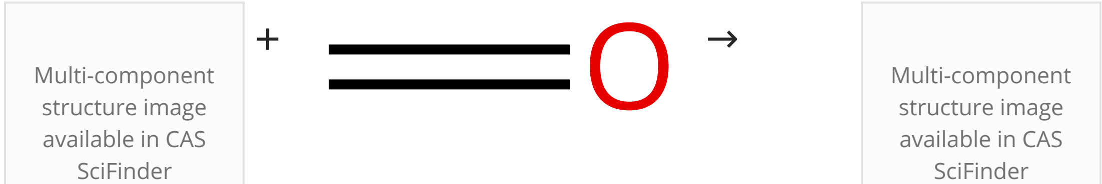
31-313-CAS-4120735	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium triacetoxyborohydride; rt

Synthesis and evaluation of a new series of Neurop eptide S receptor antagonists

By: Melamed, Jeffrey Y.; et al

Bioorganic & Medicinal Chemistry Letters (2010), 20(15), 4700-4703.

Scheme 16 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (188)

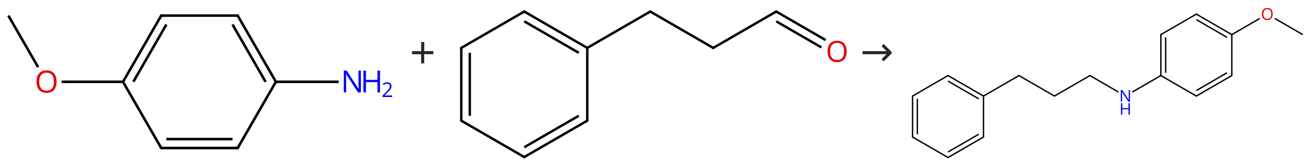
31-313-CAS-7278989	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium triacetoxyborohydride **Solvents:** *N*-Methyl-2-pyrrolidone; 16 h, 70 °C
Experimental Protocols	

Selective Transformation of a Crown Ether/sec-Ammonium Salt-Type Rotaxane to N-Alkylated Rotaxanes

By: Suzuki, Sakiko; et al

Organic Letters (2010), 12(4), 712-715.

Scheme 17 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (83) Suppliers (90) Suppliers (16)

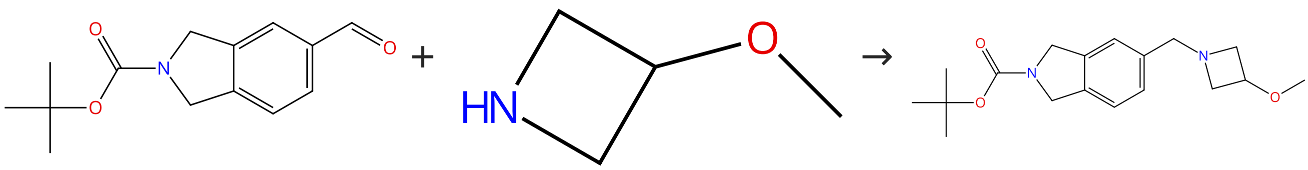
31-313-CAS-20328461	Steps: **1** Yield: **100%**
1.1	**Reagents:** Titanium isopropoxide **Solvents:** Tetrahydrofuran; 3 h, rt
1.2	**Reagents:** Sodium borohydride **Solvents:** Methanol; 4 h, reflux

Hitchhiker's Guide to Reductive Amination

By: Podyacheva, Evgeniya; et al

Synthesis (2019), 51(13), 2667-2677.

Scheme 18 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (52)

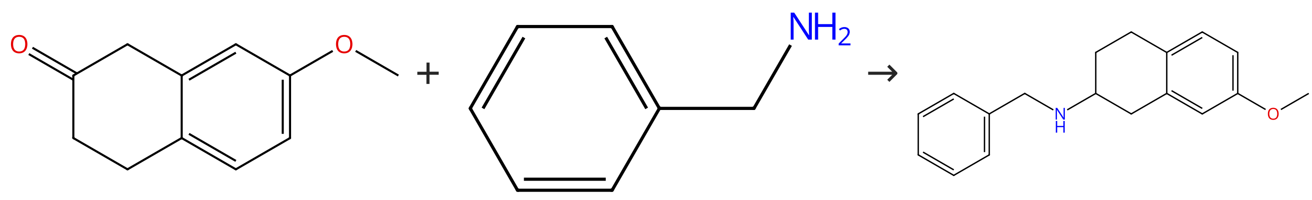
31-313-CAS-2575582	Steps: **1** Yield: **100%**
1.1	**Reagents:** Acetic acid, Sodium triacetoxyborohydride **Solvents:** Tetrahydrofuran, Water; 0.5 h, rt; 18 h, rt
1.2	**Reagents:** Ammonia **Solvents:** Water; pH 11, rt

Synthesis of substituted 5-aminomethyl tetrahydro-isoquinolines and dihydro-isoindoles

By: Fray, M. Jonathan; et al

Tetrahedron (2006), 62(29), 6869-6875.

Scheme 19 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (71) Suppliers (81) Suppliers (9)

31-313-CAS-12534793 Steps: **1** Yield: **100%**

1.1 **Solvents:** Dichloromethane; 15 min, rt

1.2 **Reagents:** Acetic acid, Sodium triacetoxyborohydride; 15 h, rt

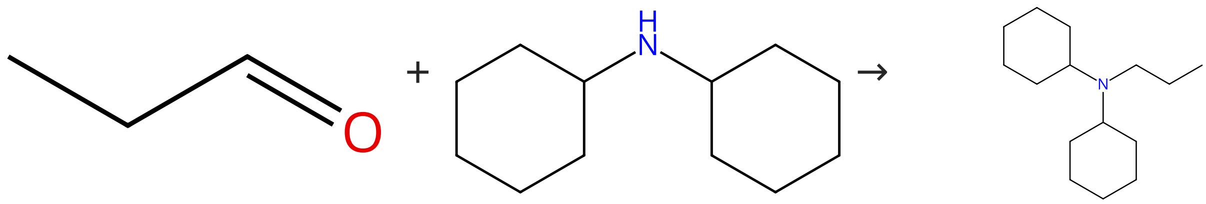
Experimental Protocols

Design and Optimization of Potent and Orally Bioavailable Tetrahydronaphthalene Raf Inhibitors

By: Gould, Alexandra E.; et al

Journal of Medicinal Chemistry (2011), 54(6), 1836-1846.

Scheme 20 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (75) Suppliers (83)

31-313-CAS-10890561 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane; rt

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

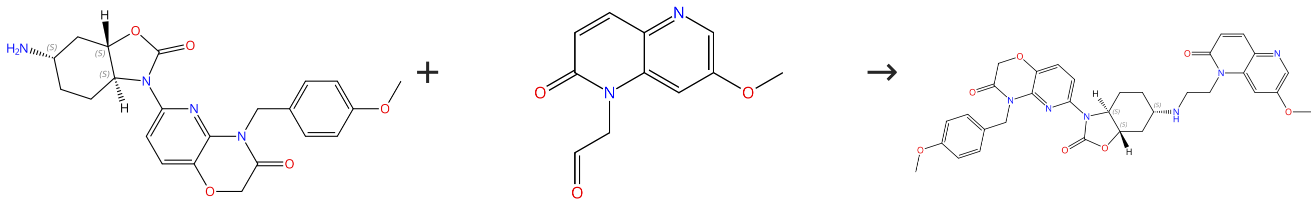
Experimental Protocols

Photoredox-Mediated C-H Functionalization and Coupling of Tertiary Aliphatic Amines with 2-Chloroazoles

By: Singh, Anuradha; et al

Organic Letters (2013), 15(20), 5390-5393.

Scheme 21 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Suppliers (3)

Absolute stereochemistry shown

31-614-CAS-34196346 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: Dichloromethane; overnight, rt

1.2 **Reagents:** Sodium carbonate
Solvents: Water; rt

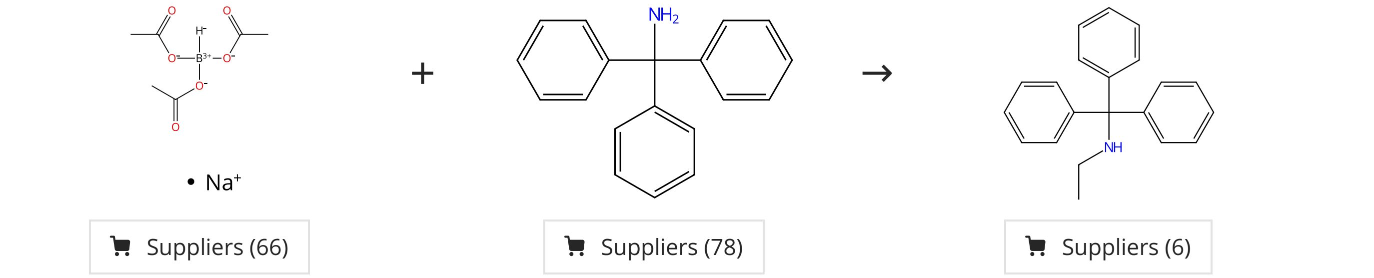
Experimental Protocols

Discovery and structure-activity relationships of a novel oxazolidinone class of bacterial type II topoisomerase inhibitors

By: Lyons, Amanda; et al

Bioorganic & Medicinal Chemistry Letters (2022), 65, 128648.

Scheme 22 (1 Reaction) Steps: **1** Yield: **100%**



31-313-CAS-22266351 Steps: **1** Yield: **100%**

1.1 **Solvents:** Acetonitrile; rt

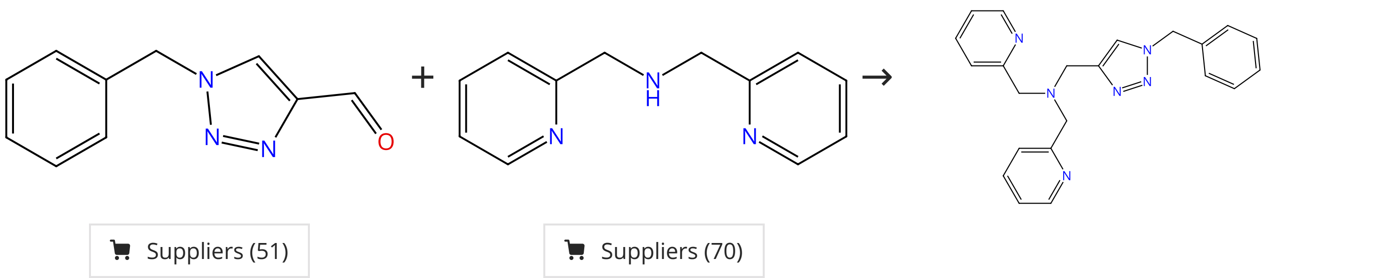
Experimental Protocols

New approach for induction of alkyl moiety to aliphatic amines by NaBH(OAc)3 with carboxylic acid

By: Tamura, Satoru; et al

Tetrahedron Letters (2020), 61(22), 151919.

Scheme 23 (1 Reaction) Steps: **1** Yield: **100%**



31-313-CAS-8385593 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane; overnight, rt

1.2 **Reagents:** Sodium bicarbonate, Sodium hydroxide
Solvents: Dichloromethane, Water; pH 8 - 9, rt

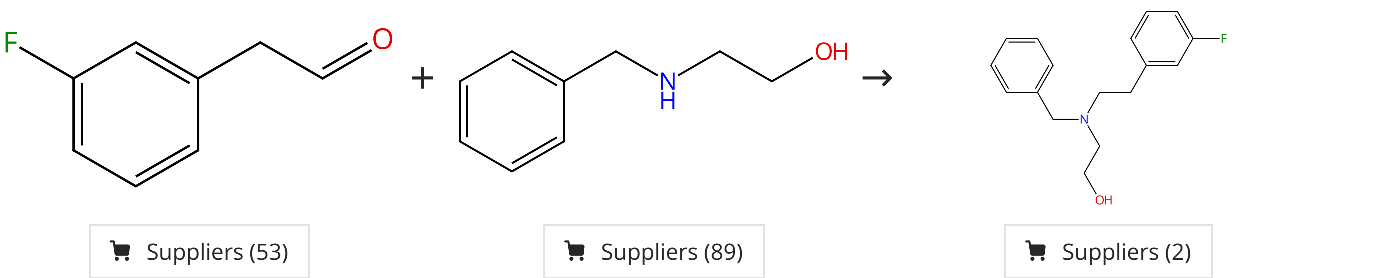
Experimental Protocols

Tailored Ligand Acceleration of the Cu-Catalyzed Azide-Alkyne Cycloaddition Reaction: Practical and Mechanistic Implications

By: Presolski, Stanislav I.; et al

Journal of the American Chemical Society (2010), 132(41), 14570-14576.

Scheme 24 (1 Reaction) Steps: **1** Yield: **100%**



31-313-CAS-11782140 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane; 14 h, rt

Experimental Protocols

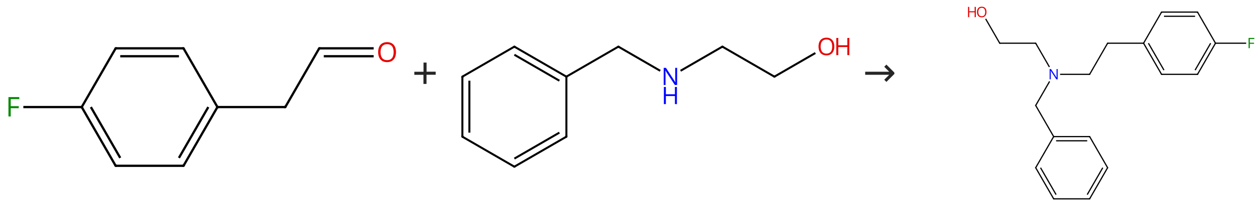
Discovery of Highly Potent and Selective Inhibitors of Neuronal Nitric Oxide Synthase by Fragment Hopping

By: Ji, Haitao; et al

Journal of Medicinal Chemistry (2009), 52(3), 779-797.

Scheme 25 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (64)

Suppliers (89)

Supplier (1)

31-313-CAS-9652841

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane; 14 h, rt

Experimental Protocols

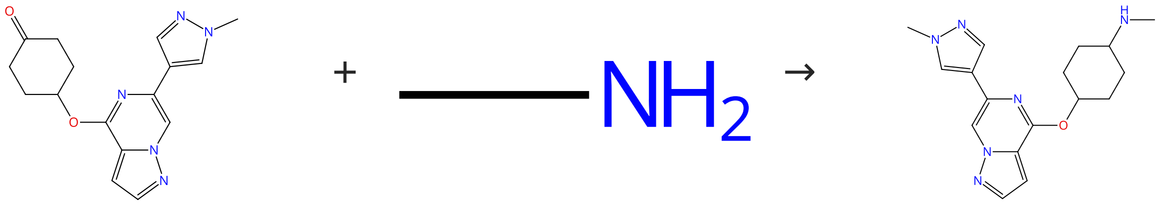
Discovery of Highly Potent and Selective Inhibitors of Neuronal Nitric Oxide Synthase by Fragment Hopping

By: Ji, Haitao; et al

Journal of Medicinal Chemistry (2009), 52(3), 779-797.

Scheme 26 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (115)

31-614-CAS-40236718

Steps: **1** Yield: **100%**

1.1 **Solvents:** Acetic acid, Tetrahydrofuran; 15 min, rt

1.2 **Reagents:** Sodium triacetoxyborohydride; overnight, rt

Experimental Protocols

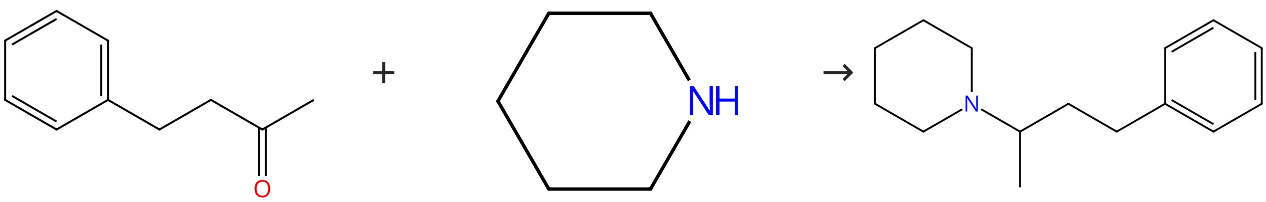
Discovery and Preclinical Characterization of BIIB129, a Covalent, Selective, and Brain-Penetrant BTK Inhibitor for the Treatment of Multiple Sclerosis

By: Himmelbauer, Martin K.; et al

Journal of Medicinal Chemistry (2024), 67(10), 8122-8140.

Scheme 27 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (77)

Suppliers (54)

31-313-CAS-20328464

Steps: **1** Yield: **100%**

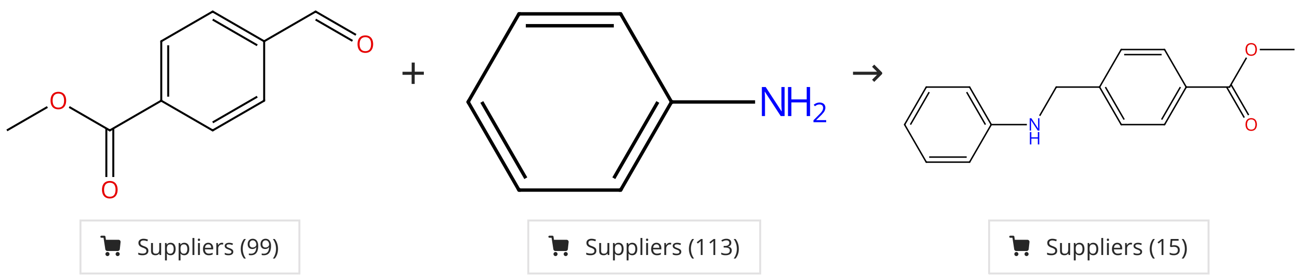
1.1 **Reagents:** Carbon monoxide
Catalysts: Dirhodium tetraacetate
Solvents: Tetrahydrofuran; 3 atm; 22 h, 50 atm, 120 °C

Hitchhiker's Guide to Reductive Amination

By: Podyacheva, Evgeniya; et al

Synthesis (2019), 51(13), 2667-2677.

Scheme 28 (1 Reaction) Steps: **1** Yield: **100%**



31-614-CAS-32317202 Steps: **1** Yield: **100%**

1.1 **Reagents:** Acetic acid
Solvents: Dichloromethane; 64 h, rt

1.2 **Reagents:** Sodium triacetoxyborohydride; 2 h, rt

1.3 **Reagents:** Water; rt

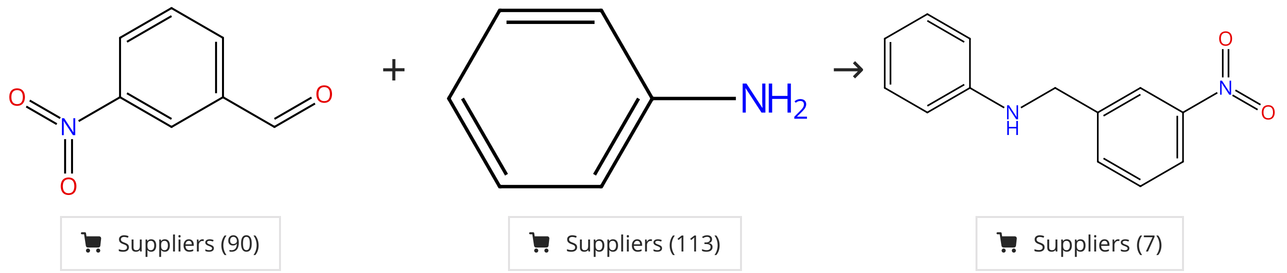
Experimental Protocols

Discovery of M_3 Antagonist-PDE4 Inhibitor Dual Pharmacology Molecules for the Treatment of Chronic Obstructive Pulmonary Disease

By: Armani, Elisabetta; et al

Journal of Medicinal Chemistry (2021), 64(13), 9100-9119.

Scheme 29 (1 Reaction) Steps: **1** Yield: **100%**



31-313-CAS-20328455 Steps: **1** Yield: **100%**

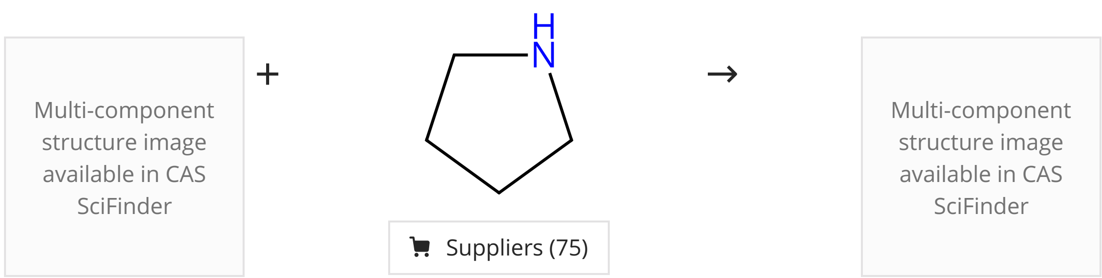
1.1 **Reagents:** Acetic acid, Sodium cyanoborohydride
Solvents: Methanol; overnight, rt

Hitchhiker's Guide to Reductive Amination

By: Podyacheva, Evgeniya; et al

Synthesis (2019), 51(13), 2667-2677.

Scheme 30 (1 Reaction) Steps: **1** Yield: **100%**



31-313-CAS-12654461 Steps: **1** Yield: **100%**

1.1 **Reagents:** Formic acid
Solvents: Dimethylformamide; 5 h, 70 °C

1.2 **Reagents:** Potassium formate
Catalysts: Palladium diacetate; 4 h, 70 °C; 70 °C → rt

1.3 **Reagents:** Potassium bifluoride
Solvents: Water; 20 min, rt

Experimental Protocols

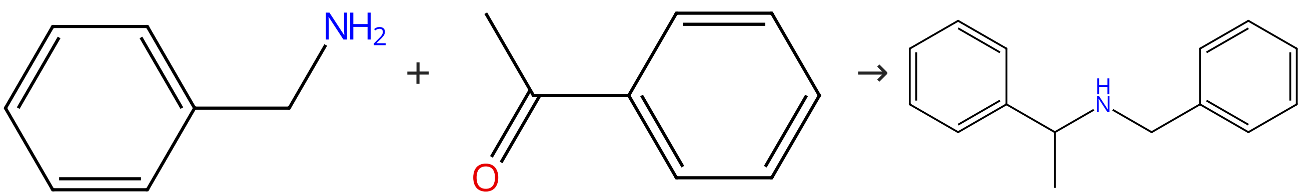
Functionalization of Organotrifluoroborates: Reductive Amination

By: Molander, Gary A.; et al

Journal of Organic Chemistry (2008), 73(10), 3885-3891.

Scheme 31 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (81) Suppliers (105) Suppliers (53)

31-313-CAS-20328458	Steps: **1** Yield: **100%**
1.1	**Reagents:** Titanium isopropoxide **Solvents:** Tetrahydrofuran; 3 h, rt
1.2	**Reagents:** Sodium borohydride **Solvents:** Methanol; 4 h, reflux

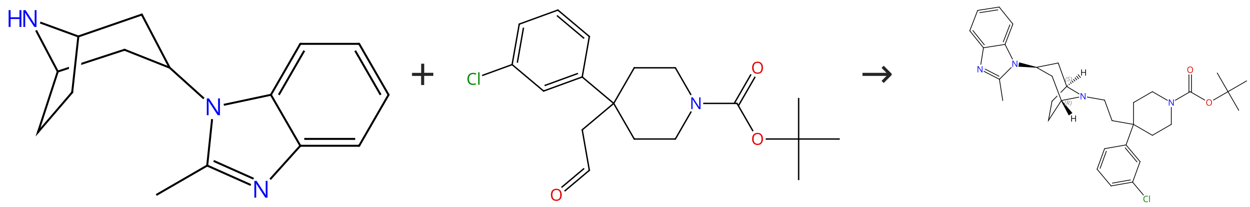
Hitchhiker's Guide to Reductive Amination

By: Podyacheva, Evgeniya; et al

Synthesis (2019), 51(13), 2667-2677.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (36) Suppliers (3)

Relative stereochemistry shown

31-313-CAS-81623	Steps: **1** Yield: **99%**
1.1	**Reagents:** Triethylamine, Sodium triacetoxyborohydride **Solvents:** Dichloromethane; 1 h, rt
1.2	**Reagents:** Sodium bicarbonate **Solvents:** Water

Experimental Protocols

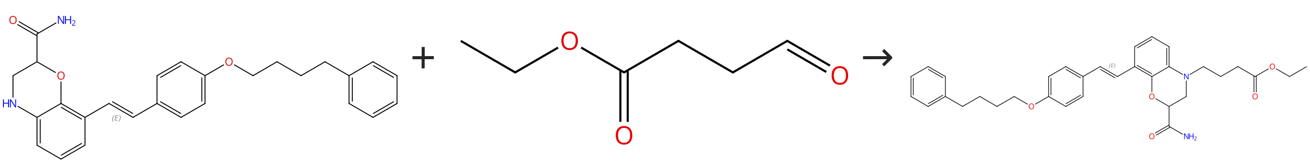
Discovery of Bioavailable 4,4-Disubstituted Piperidines as Potent Ligands of the Chemokine Receptor 5 and Inhibitors of the Human Immunodeficiency Virus-1

By: Kazmierski, Wieslaw M.; et al

Journal of Medicinal Chemistry (2008), 51(20), 6538-6546.

Scheme 33 (1 Reaction)

Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (47)

Double bond geometry shown

31-313-CAS-2857704	Steps: **1** Yield: **99%**
1.1	**Reagents:** Acetic acid, Sodium triacetoxyborohydride **Solvents:** Tetrahydrofuran; 0 °C; 3 h, rt

Experimental Protocols

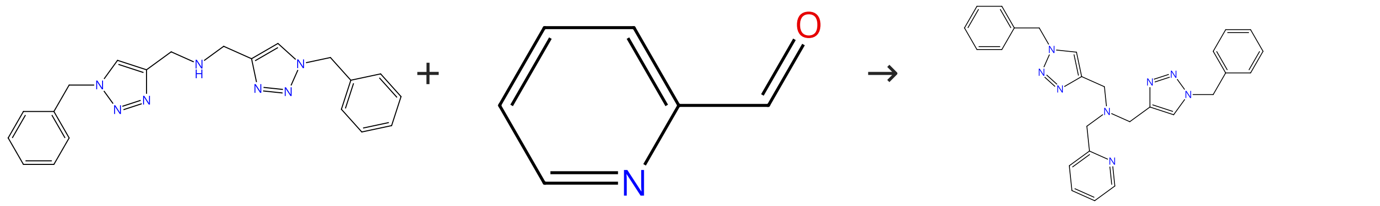
Discovery of a potent, orally available dual CysLT$_1$ and CysLT$_2$ antagonist with dicarboxylic acid

By: Itadani, Satoshi; et al

Bioorganic & Medicinal Chemistry (2015), 23(9), 2079-2097.

Scheme 34 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (86)

Supplier (1)

31-313-CAS-4082097

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane; overnight, rt

1.2 **Reagents:** Sodium bicarbonate, Sodium hydroxide
Solvents: Dichloromethane, Water; pH 8 - 9, rt

Experimental Protocols

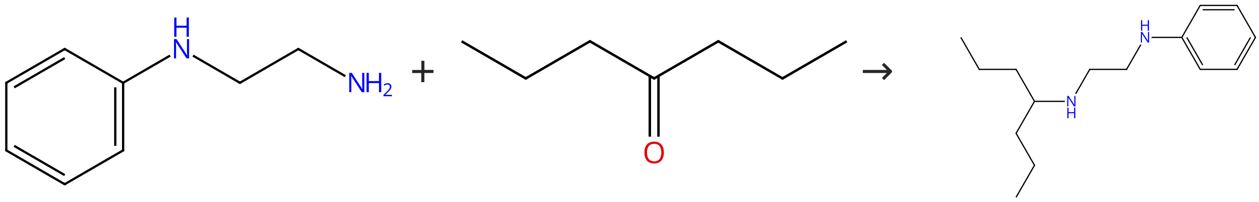
Tailored Ligand Acceleration of the Cu-Catalyzed Azide-Alkyne Cycloaddition Reaction: Practical and Mechan istic Implic ations

By: Presolski, Stanislav I.; et al

Journal of the American Chemical Society (2010), 132(41), 14570-14576.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (67)

Suppliers (62)

31-313-CAS-4175742

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

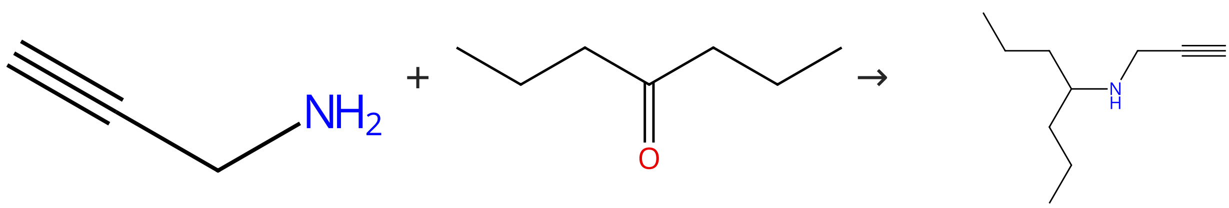
Reductive Amination of Aldehydes and Ketones with Sodium Triacetoxyborohydride. Studies on Direct and Indirect Reductive Amination Procedures

By: Abdel-Magid, Ahmed F.; et al

Journal of Organic Chemistry (1996), 61(11), 3849-3862.

Scheme 36 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (75)

Suppliers (62)

Suppliers (12)

31-313-CAS-1801073

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

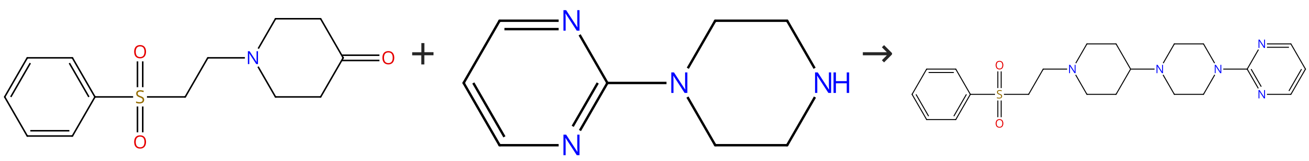
Reductive Amination of Aldehydes and Ketones with Sodium Triacetoxyborohydride. Studies on Direct and Indirect Reductive Amination Procedures

By: Abdel-Magid, Ahmed F.; et al

Journal of Organic Chemistry (1996), 61(11), 3849-3862.

Scheme 37 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (2) | Suppliers (101) | Suppliers (2)

31-313-CAS-9837211

Steps: **1** Yield: **99%**

1.1 **Reagents:** Acetic acid, Sodium triacetoxyborohydride
Solvents: Dichloromethane, Water; 12 - 15 h, rt

1.2 **Reagents:** Sodium bicarbonate
Solvents: Dichloromethane, Water; rt

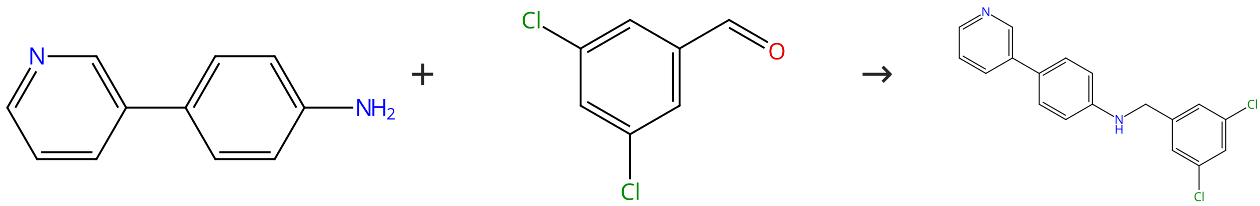
Synthesis of hetero-aryl-piperazines and hetero-aryl-bipiperidines with a restricted side chain and their affinities for 5- HT$_{1A}$ receptor

By: Yoo, Kyung Ho; et al

Archiv der Pharmazie (Weinheim, Germany) (2003), 336(4-5), 208-215.

Scheme 38 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (71) | Suppliers (86)

31-614-CAS-31497183

Steps: **1** Yield: **99%**

1.1 **Reagents:** Acetic acid
Solvents: Dichloromethane; 30 min, 23 °C

1.2 **Reagents:** Sodium triacetoxyborohydride; 18 h, rt

Experimental Protocols

Structure-based optimization of ML300-derived, noncovalent inhibitors targeting the severe acute respiratory syndrome coronavirus 3CL protease (SARS-CoV-2 3CL^{pro})

By: Han, Sang Hoon; et al

Journal of Medicinal Chemistry (2022), 65(4), 2880-2904.

Scheme 39 (1 Reaction)

Steps: **1** Yield: **99%**

Suppliers (3) | Suppliers (115) | Suppliers (2)

31-313-CAS-15048009

Steps: **1** Yield: **99%**

1.1 **Solvents:** Methanol; rt; 6 h, reflux; reflux → - 20 °C

1.2 **Reagents:** Sodium borohydride; 1 h, -20 °C; 3 h, rt

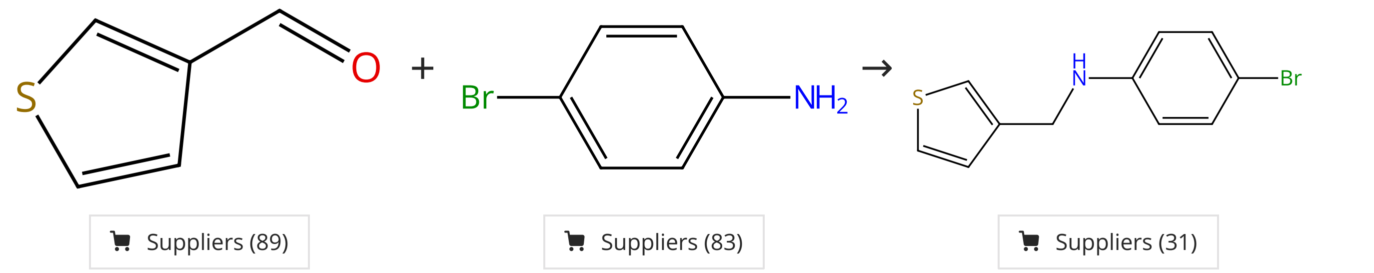
Experimental Protocols

Discovery of Highly Potent and Selective Inhibitors of Neuronal Nitric Oxide Synthase by Fragment Hopping

By: Ji, Haitao; et al

Journal of Medicinal Chemistry (2009), 52(3), 779-797.

Scheme 40 (1 Reaction) Steps: **1** Yield: **99%**



31-614-CAS-31320948	Steps: **1** Yield: **99%**
1.1	**Reagents:** Sodium triacetoxyborohydride **Solvents:** 1,2-Dichloroethane; 1 h, rt

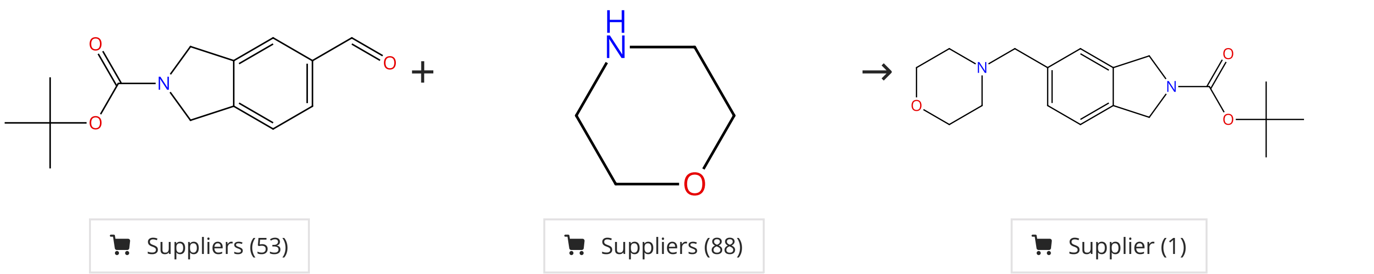
Experimental Protocols

Structure-based optimization of ML300-derived, noncovalent inhibitors targeting the severe acute respiratory syndrome coronavirus 3CL protease (SARS-CoV-2 3CL^{pro})

By: Han, Sang Hoon; et al

Journal of Medicinal Chemistry (2022), 65(4), 2880-2904.

Scheme 41 (1 Reaction) Steps: **1** Yield: **99%**



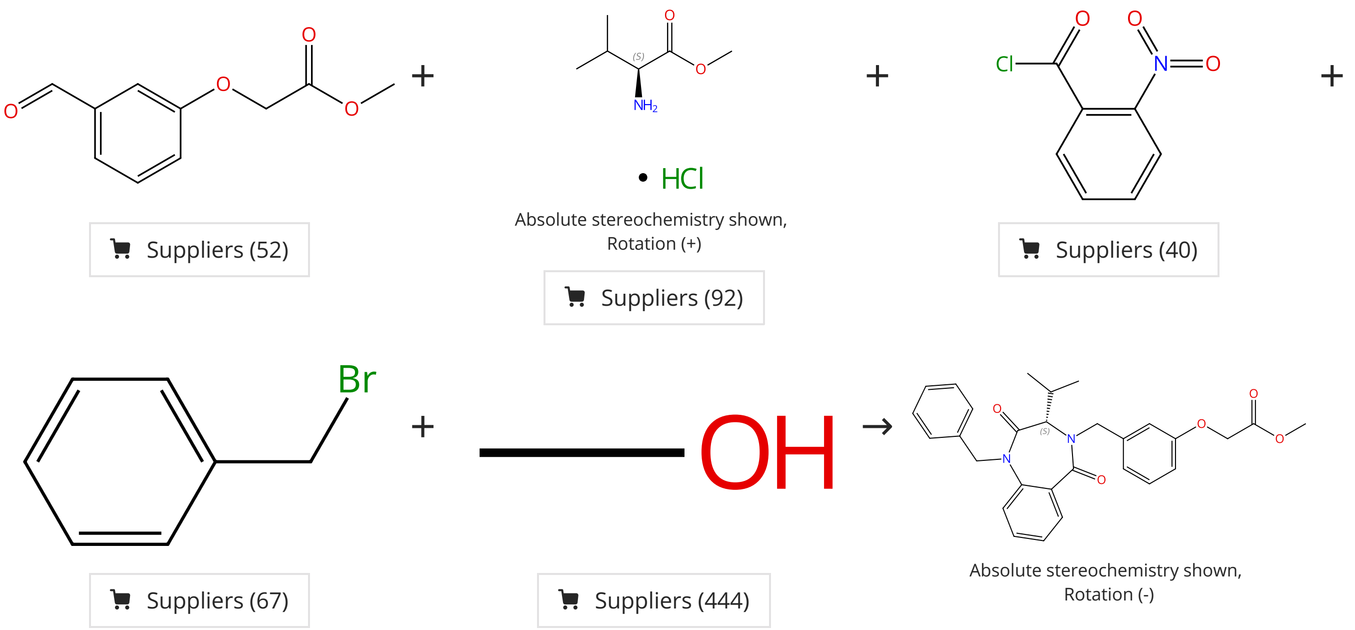
31-313-CAS-13827009	Steps: **1** Yield: **99%**
1.1	**Reagents:** Acetic acid, Sodium triacetoxyborohydride **Solvents:** Tetrahydrofuran, Water; 0.5 h, rt; 18 h, rt
1.2	**Reagents:** Ammonia **Solvents:** Water; pH 11, rt

Synthesis of substituted 5-aminomethyl tetrahydro-isoquin olines and dihydro-isoindoles

By: Fray, M. Jonathan; et al

Tetrahedron (2006), 62(29), 6869-6875.

Scheme 42 (1 Reaction) Steps: **1** Yield: **99%**



31-031-CAS-7809569	Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, 4-(Dimethylamino) pyridine, Polyethylene glycol monomethyl ether
Solvents: Acetonitrile; 23 h, rt

1.2 **Reagents:** Sodium acetate, Sodium triacetoxyborohydride
Solvents: Dichloromethane; 5 h, rt

1.3 **Reagents:** Tetrabutylammonium iodide, Potassium carbonate
Solvents: Dichloromethane; 5 h, rt

1.4 **Reagents:** Acetic acid, Zinc; 2 h, rt

1.5 **Reagents:** Trifluoroacetic acid, Sodium trifluoroacetate
Solvents: Acetonitrile; 13 h, rt

1.6 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 2 h, rt

1.7 **Reagents:** Sodium carbonate
Solvents: Methanol; 5 min, rt

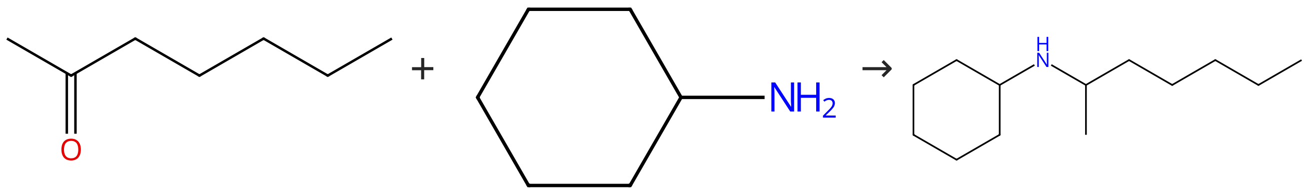
Experimental Protocols

Liquid-Phase Combinatorial Synthesis of 1,4-Benzodiazepine-2,5-diones as the Candidates of Endothelin Receptor Antagonism

By: Cheng, Ming-Fu; et al

Journal of Combinatorial Chemistry (2004), 6(1), 99-104.

Scheme 43 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (68) Suppliers (85) Suppliers (18)

31-313-CAS-7097853	Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: Tetrahydrofuran

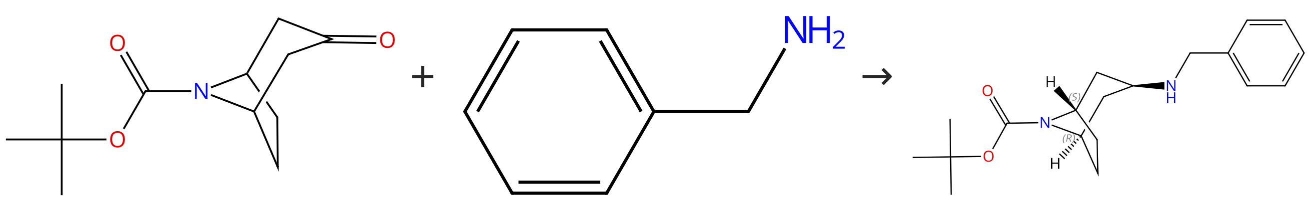
1.2 **Reagents:** Sodium hydroxide
Solvents: Water

Reductive Amination of Aldehydes and Ketones with Sodium Triacetoxyborohydride. Studies on Direct and Indirect Reductive Amination Procedures

By: Abdel-Magid, Ahmed F.; et al

Journal of Organic Chemistry (1996), 61(11), 3849-3862.

Scheme 44 (1 Reaction) Steps: **1** Yield: **99%**



Relative stereochemistry shown

Suppliers (104) Suppliers (81) Suppliers (9)

31-313-CAS-15282046	Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: Dichloromethane; 45 min, rt; 2 d, rt

1.2 **Reagents:** Sodium carbonate
Solvents: Water; 1 h, rt

Experimental Protocols

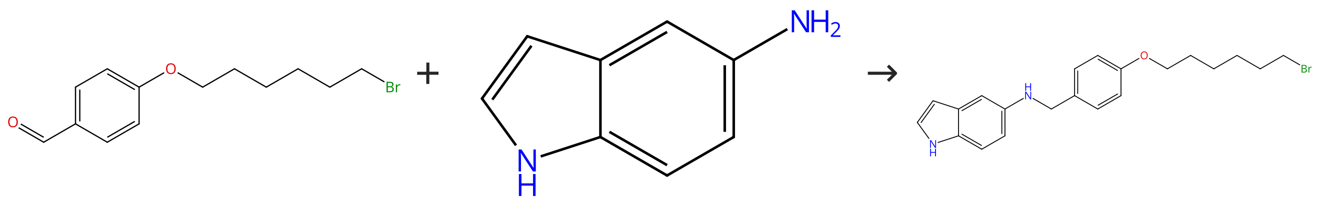
Discovery of Bioavailable 4,4-Disubstituted Piperidines as Potent Ligands of the Chemokine Receptor 5 and Inhibitors of the Human Immunodeficiency Virus-1

By: Kazmierski, Wieslaw M.; et al

Journal of Medicinal Chemistry (2008), 51(20), 6538-6546.

Scheme 45 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (38)

Suppliers (95)

31-313-CAS-22782675

Steps: **1** Yield: **99%**

1.1 **Solvents:** 1,2-Dichloroethane; 30 min, rt
1.2 **Reagents:** Sodium triacetoxyborohydride; 24 h, rt
1.3 **Reagents:** Sodium bicarbonate
Solvents: Water; rt

Experimental Protocols

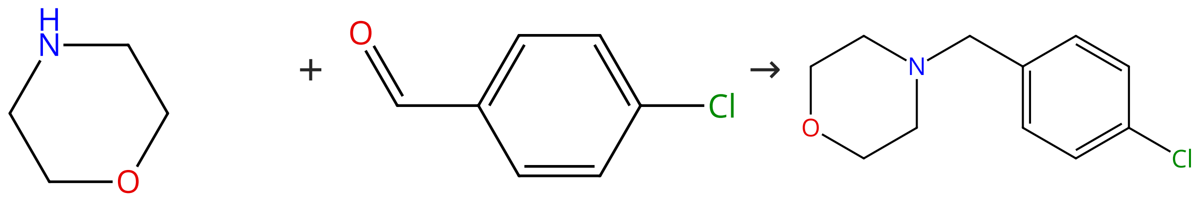
Galantamine derivatives as acetylcholinesterase inhibitors: docking, design, synthesis, and inhibitory activity

By: Doytchinova, Irini; et al

Neuromethods (2018), 132, 163-176.

Scheme 46 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (88)

Suppliers (99)

Suppliers (40)

31-313-CAS-20716673

Steps: **1** Yield: **99%**

1.1 **Solvents:** Dimethylacetamide; 1 h, rt; rt → 10 °C
1.2 **Reagents:** Acetic acid; 10 °C
1.3 **Reagents:** Sodium triacetoxyborohydride; 16 - 18 h, 10 °C
1.4 **Solvents:** Water; 10 °C

Experimental Protocols

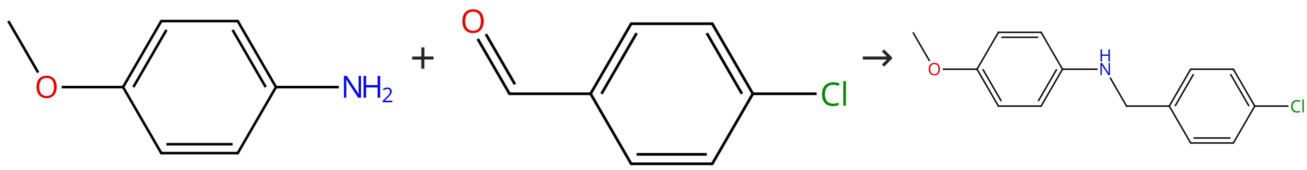
A Quantitative Assay of Sodium Triacetoxyborohydride

By: Zacuto, Michael J.; et al

Organic Process Research & Development (2019), 23(9), 2080-2087.

Scheme 47 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (83)

Suppliers (99)

Suppliers (41)

31-313-CAS-20328448

Steps: **1** Yield: **99%**

1.1 **Solvents:** 1,2-Dichloroethane; 15 min, rt
1.2 **Reagents:** Sodium triacetoxyborohydride; 18 h, rt
1.3 **Reagents:** Sodium bicarbonate
Solvents: Water

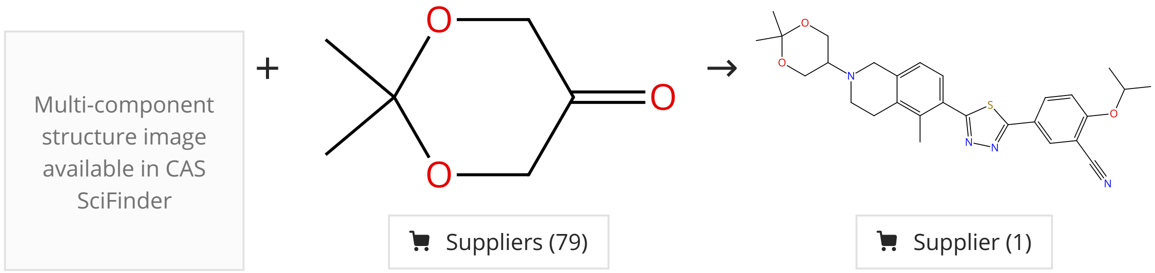
Hitchhiker's Guide to Reductive Amination

By: Podyacheva, Evgeniya; et al

Synthesis (2019), 51(13), 2667-2677.

Scheme 48 (1 Reaction)

Steps: **1** Yield: **99%**



31-313-CAS-9778289

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: Dichloromethane; 18 h, rt

Experimental Protocols

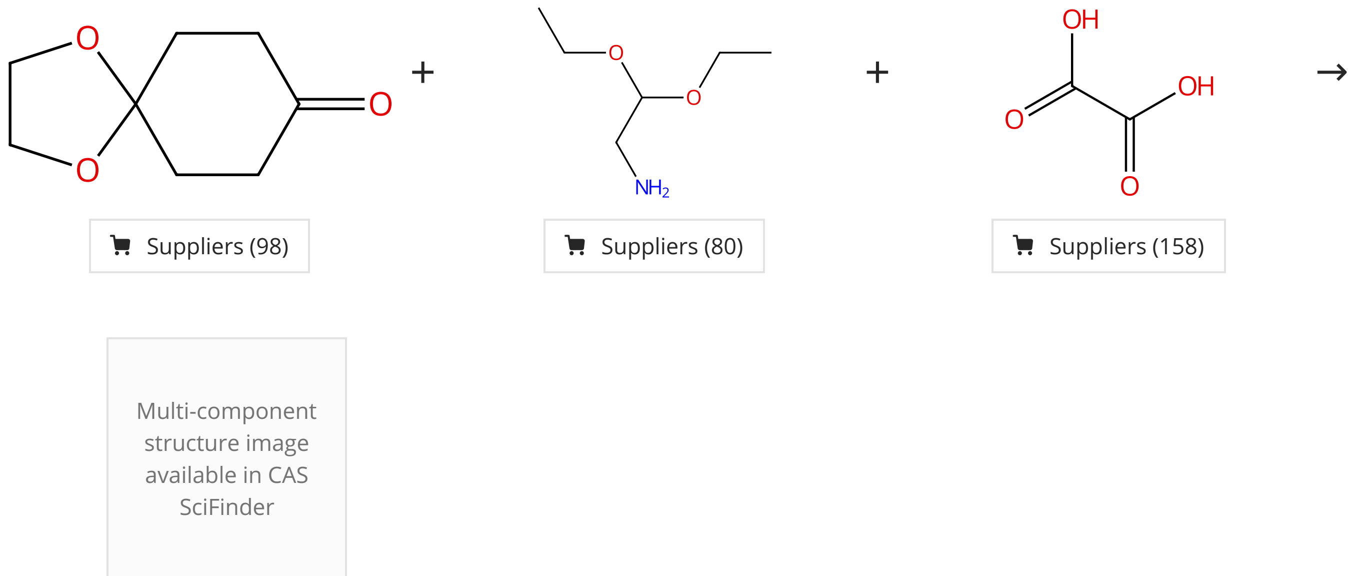
Navigating CYP1A Induction and Arylhydrocarbon Receptor Agonism in Drug Discovery. A Case History with S1P$_1$ Agonists

By: Taylor, Simon J.; et al

Journal of Medicinal Chemistry (2015), 58(20), 8236-8256.

Scheme 49 (1 Reaction)

Steps: **1** Yield: **99%**



31-313-CAS-6671356

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: 1,2-Dichloroethane

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

1.3 **Solvents:** Methanol

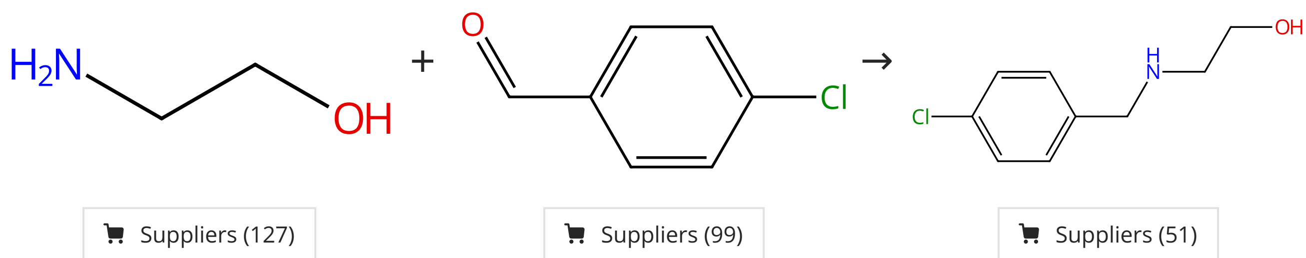
Reductive Amination of Aldehydes and Ketones with Sodium Triacetoxyborohydride. Studies on Direct and Indirect Reductive Amination Procedures

By: Abdel-Magid, Ahmed F.; et al

Journal of Organic Chemistry (1996), 61(11), 3849-3862.

Scheme 50 (1 Reaction)

Steps: **1** Yield: **99%**



31-313-CAS-14239943 Steps: **1** Yield: **99%**	**Discovery of Highly Potent and Selective Inhibitors of Neuronal Nitric Oxide Synthase by Fragment Hopping**
1.1 **Solvents:** Ethanol; overnight, 60 °C 1.2 **Reagents:** Sodium borohydride **Solvents:** Methanol; cooled; overnight, rt	By: Ji, Haitao; et al Journal of Medicinal Chemistry (2009), 52(3), 779-797.
Experimental Protocols	